

Class 14: RNASeq mini project

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Background

Data Import

Reading the counts and metadata CSV file

```
counts <- read.csv("GSE37704_featurecounts.csv", row.names = 1)
metadata <- read.csv("GSE37704_metadata.csv")
```

```
head(counts)
```

	length	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370
ENSG00000186092	918	0	0	0	0	0
ENSG00000279928	718	0	0	0	0	0
ENSG00000279457	1982	23	28	29	29	28
ENSG00000278566	939	0	0	0	0	0
ENSG00000273547	939	0	0	0	0	0

ENSG00000187634	3214	124	123	205	207	212
	SRR493371					
ENSG00000186092	0					
ENSG00000279928	0					
ENSG00000279457	46					
ENSG00000278566	0					
ENSG00000273547	0					
ENSG00000187634	258					

```
metadata
```

	id	condition
1	SRR493366	control_sirna
2	SRR493367	control_sirna
3	SRR493368	control_sirna
4	SRR493369	hoxa1_kd
5	SRR493370	hoxa1_kd
6	SRR493371	hoxa1_kd

```
ncol(counts)
```

```
[1] 7
```

```
nrow(metadata)
```

```
[1] 6
```

Looks like we need to get rid of the first “length” column of our countsobject.

```
cleancounts <- counts[,-1]
head(cleancounts)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000186092	0	0	0	0	0	0
ENSG00000279928	0	0	0	0	0	0
ENSG00000279457	23	28	29	29	28	46
ENSG00000278566	0	0	0	0	0	0
ENSG00000273547	0	0	0	0	0	0
ENSG00000187634	124	123	205	207	212	258

```
colnames(cleancounts)
```

```
[1] "SRR493366" "SRR493367" "SRR493368" "SRR493369" "SRR493370" "SRR493371"
```

```
colnames(cleancounts) == metadata$id
```

```
[1] TRUE TRUE TRUE TRUE TRUE TRUE
```

Remove zero count genes

There are lots of genes with zero counts. We can remove these from further analysis.

```
head(cleancounts)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000186092	0	0	0	0	0	0
ENSG00000279928	0	0	0	0	0	0
ENSG00000279457	23	28	29	29	28	46
ENSG00000278566	0	0	0	0	0	0
ENSG00000273547	0	0	0	0	0	0
ENSG00000187634	124	123	205	207	212	258

```
to.keep.inds <- rowSums(cleancounts) > 0  
nonzero_counts <- cleancounts[to.keep.inds,]
```

DESeq analysis

```
library(DESeq2)
```

Warning: package 'matrixStats' was built under R version 4.5.2

```
dds <- DESeqDataSetFromMatrix(countData= nonzero_counts,  
                              colData= metadata,  
                              design= ~condition)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

Get Results

```
dds <- DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

```
res <- results(dds)
```

```
res
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 15975 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.9136	0.1792571	0.3248216	0.551863	5.81042e-01
ENSG00000187634	183.2296	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.1881	-0.6927205	0.0548465	-12.630158	1.43990e-36
ENSG00000187961	209.6379	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.2551	0.0405765	0.2718928	0.149237	8.81366e-01
...	...	...	...	...	...
ENSG00000273748	35.30265	0.674387	0.303666	2.220817	2.63633e-02
ENSG00000278817	2.42302	-0.388988	1.130394	-0.344117	7.30758e-01
ENSG00000278384	1.10180	0.332991	1.660261	0.200565	8.41039e-01
ENSG00000276345	73.64496	-0.356181	0.207716	-1.714752	8.63908e-02
ENSG00000271254	181.59590	-0.609667	0.141320	-4.314071	1.60276e-05
	padj				
	<numeric>				
ENSG00000279457	6.86555e-01				
ENSG00000187634	5.15718e-03				
ENSG00000188976	1.76549e-35				

```

ENSG000000187961 1.13413e-07
ENSG000000187583 9.19031e-01
...
ENSG000000273748 4.79091e-02
ENSG000000278817 8.09772e-01
ENSG000000278384 8.92654e-01
ENSG000000276345 1.39762e-01
ENSG000000271254 4.53648e-05

```

Data Visualization

Volcano plot

```

library(ggplot2)

res$color <- "gray" # default
res$color[abs(res$log2FoldChange) > 2] <- "red"
res$color[(res$padj < 0.01) & (abs(res$log2FoldChange) > 2)] <- "blue"

# Create a volcano plot with ggplot2
ggplot(res, aes(x = log2FoldChange, y = -log10(padj), color = color)) +
  geom_point(alpha = 0.7, size = 2) +

  # Threshold lines
  geom_vline(xintercept = c(-2, 2), color = "magenta", linewidth = 0.7) +
  geom_hline(yintercept = -log10(0.1), color = "magenta", linewidth = 0.7) +

  # Custom color scale (keep your gray-red-blue scheme)
  scale_color_identity(
    guide = "legend",
    breaks = c("gray", "red", "blue"),
    labels = c("Not significant", "|log2FC| > 2", "|log2FC| > 2 & padj < 0.01"),
    name = ""
  ) +

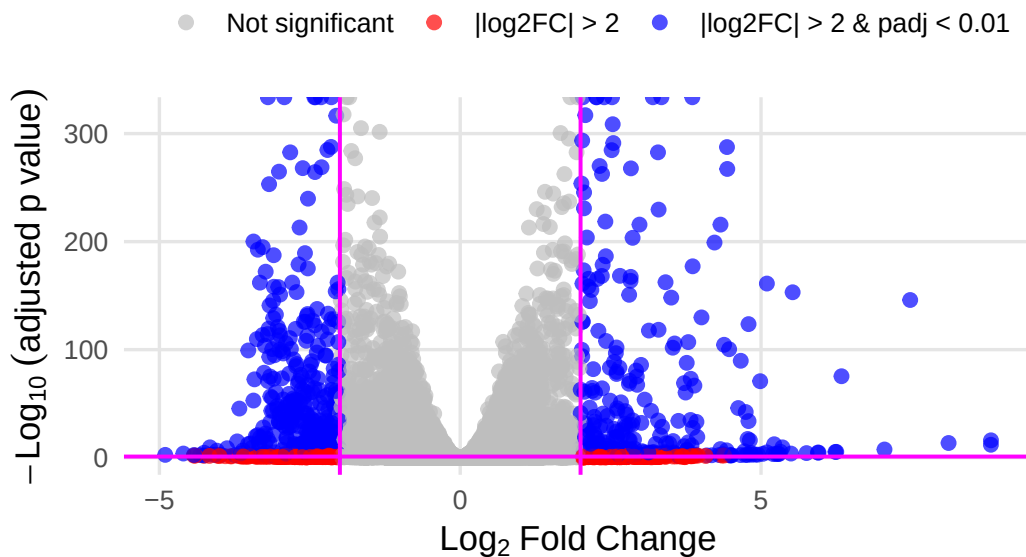
  # Labels and theme
  labs(
    title = "Volcano Plot of Differential Expression",
    x = expression(Log[2]~Fold~Change),
    y = expression(-Log[10]~(adjusted~p~value))
  ) +

```

```
theme_minimal(base_size = 13) +
theme(
  plot.title = element_text(face = "bold", size = 15, hjust = 0.5),
  legend.position = "top",
  panel.grid.minor = element_blank(),
  panel.grid.major = element_line(color = "gray90")
)
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (`geom\_point()`).

Volcano Plot of Differential Expression



Add Threshold lines for fold-change and P-value and color our subset of genes that make these threshold cut-offs in the plot.

Add Annotation

Add gene symbols and entrez ids

```
library(AnnotationDbi)
library(org.Hs.eg.db)
```

```
res$symbol <- mapIds(x = org.Hs.eg.db,
                    keys = row.names(res),
                    keytype = "ENSEMBL",
                    column = "SYMBOL")
```

'select()' returned 1:many mapping between keys and columns

```
res$entrez <- mapIds(x = org.Hs.eg.db,
                    keys = row.names(res),
                    keytype = "ENSEMBL",
                    column = "ENTREZID")
```

'select()' returned 1:many mapping between keys and columns

Pathway Analysis

Run gage analysis ### KEGG terms

```
library(gage)
```

```
library(gageData)
library(pathview)
```

```
#####
Pathview is an open source software package distributed under GNU General
Public License version 3 (GPLv3). Details of GPLv3 is available at
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
formally cite the original Pathview paper (not just mention it) in publications
or products. For details, do citation("pathview") within R.
```

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG license agreement (details at <http://www.kegg.jp/kegg/legal.html>).

```
#####
```

We need a named vector of fold-change values as input for gage

```
foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)
```

```
<NA>      148398      26155      339451      84069      84808
0.17925708 0.42645712 -0.69272046 0.72975561 0.04057653 0.54281049
```

```
data(kegg.sets.hs)
```

```
keggres = gage(foldchanges, gsets=kegg.sets.hs)
head(keggres$less, 5)
```

	p.geomean	stat.mean
hsa04110 Cell cycle	8.995727e-06	-4.378644
hsa03030 DNA replication	9.424076e-05	-3.951803
hsa05130 Pathogenic Escherichia coli infection	1.405864e-04	-3.765330
hsa03013 RNA transport	1.246882e-03	-3.059466
hsa03440 Homologous recombination	3.066756e-03	-2.852899

	p.val	q.val
hsa04110 Cell cycle	8.995727e-06	0.001889103
hsa03030 DNA replication	9.424076e-05	0.009841047
hsa05130 Pathogenic Escherichia coli infection	1.405864e-04	0.009841047
hsa03013 RNA transport	1.246882e-03	0.065461279
hsa03440 Homologous recombination	3.066756e-03	0.128803765

	set.size	exp1
hsa04110 Cell cycle	121	8.995727e-06
hsa03030 DNA replication	36	9.424076e-05
hsa05130 Pathogenic Escherichia coli infection	53	1.405864e-04
hsa03013 RNA transport	144	1.246882e-03
hsa03440 Homologous recombination	28	3.066756e-03

```
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/BIMM143/class14

Info: Writing image file hsa04110.pathview.png

Reactome

Lots of folks like the reactome web interface. You can also run this as an R function but let's look at the website first. <https://reactome.org/>

The Website wants a text file with one gene symbol per line of the genes you want to map to pathways.

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]
```

And write out a file:

```
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

Save our results

```
write.csv(res, file = "myresults.csv")
```